

15. Three vertices of a regular octagon are used as vertices of a triangle ABC . The exact smallest possible ratio k for $\frac{\text{Area of triangle}}{\text{Area of octagon}}$ can be expressed as $k = \frac{a+b\sqrt{c}}{d}$ in reduced form with a, b, c , and d integers and $d > 0$. Determine the sum $(a+b+c+d)$.
16. Determine the remainder when $(x+1)^5 + (x+2)^4 + (x+3)^3 + (x+4)^2 + (x+5)$ is divided by $(x+2)$. Express your answer as an integer or as a common or improper fraction.
17. $i = \sqrt{-1}$ and $z = a+bi$, a and b real. When graphed in the complex plane, the roots of $z^{12} = 1$ determine vertices of a regular dodecagon. The perimeter of this regular dodecagon can be expressed as $k = m\sqrt{n} + p\sqrt{q}$, m, n, p , and q integers. Determine the sum $(m+n+p+q)$.
18. A set of points in the xy -plane consists of all those points satisfying the inequality $x^2 + y^2 < 1$. The graph of this region has numeric area Z . A curve with the equation $\frac{1}{x} + \frac{1}{y} = \frac{1}{xy}$, where $xy \neq 0$, separates the region into two parts. The numeric area A of the larger part can be represented as $A = \frac{k\pi + w}{p}$ in reduced form with k, w , and p integers and $p > 0$. Determine the sum $(k+w+p)$.
19. The sum of all values for x in the interval $[0, 2\pi]$ such that $\sqrt{3} \sin x + \cos x = \sqrt{2}$ is $k\pi$. Determine the value of k . Express your answer as an integer or as a common or improper fraction.
20. The foci of a graph of an ellipse $\frac{x^2}{k} + \frac{y^2}{w} = 1$ are located at $(0, \sqrt{5})$ and $(0, -\sqrt{5})$. The major axis has length 6. Determine the sum $(k+w)$.